

EcoFuzz: Adaptive Energy-Saving Greybox Fuzzing as a Variant of the Adversarial Multi-Armed Bandit

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CGF & Markov Chain

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Implementation
EcoFuzz

Fuzzing & AFL



AFL의 문제

Seed Scheduling Algorithm

constant power schedule

High-frequency paths

해결방안 : AFLFast

AFLFast

(Markov chain)

~~transition probability~~

~~flexibly~~

trade-off

exploration vs. exploitation

American Fuzzy Lop의 기본 동작

coverage-based greybox fuzzer

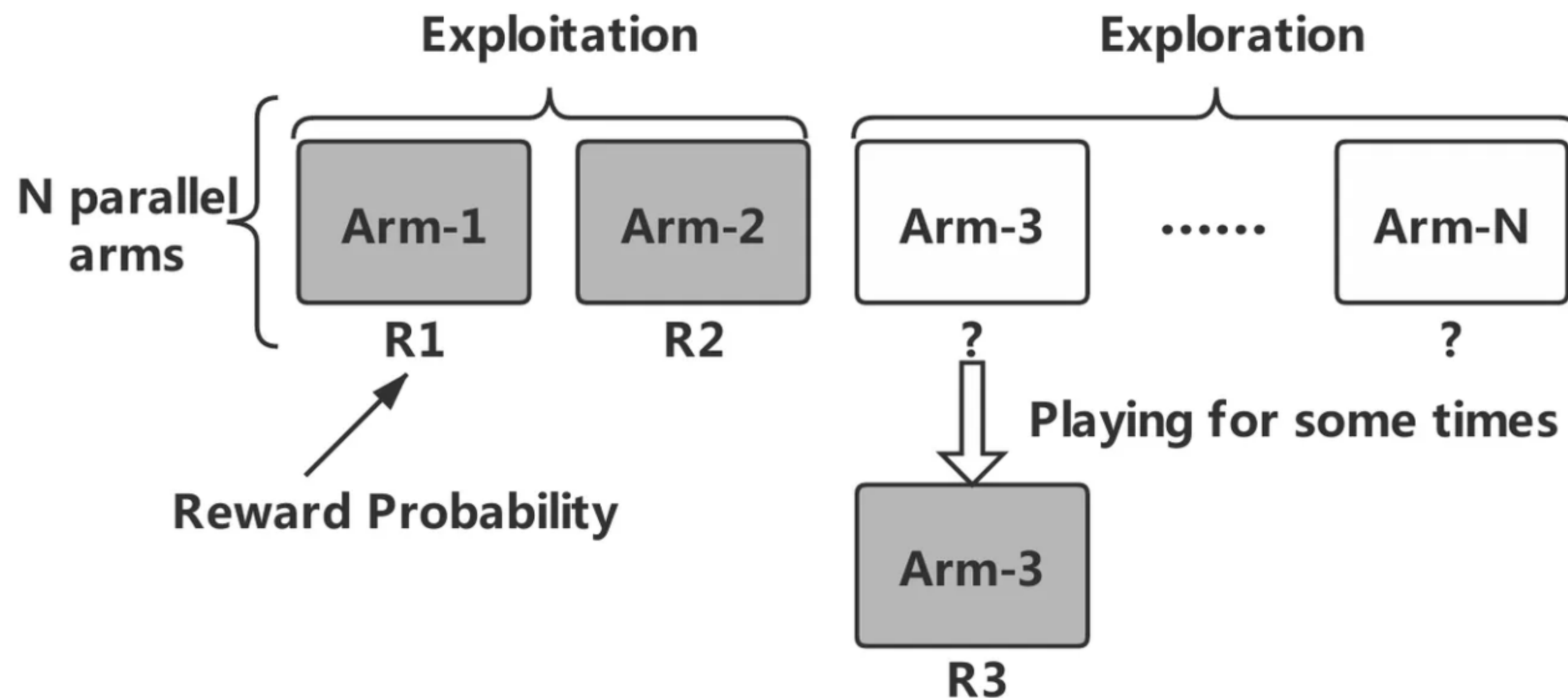
lightweight instrumentation

seed queue

coverage feedback

heuristic score

Multi-Armed Bandit(MAB)



unknown reward → exploration
maximize total reward → exploitation

Why not classic MAB?

가정

arm 수 고정

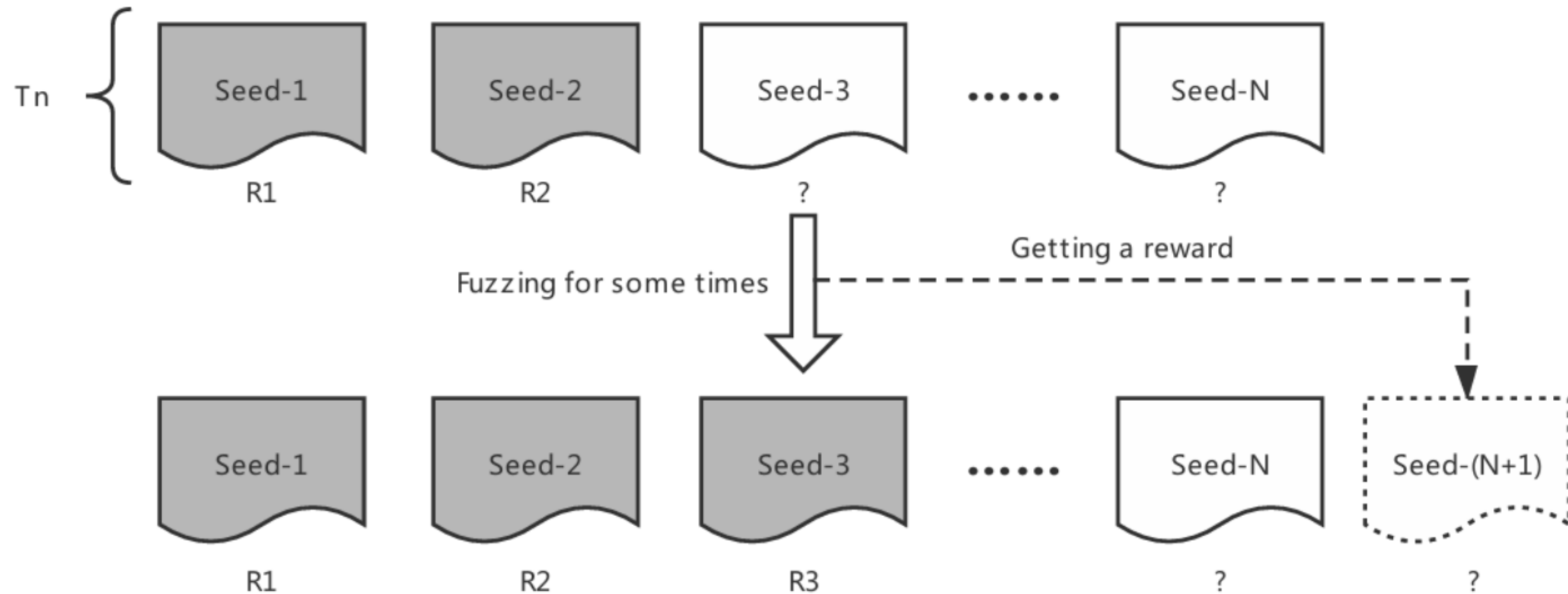
**reward distribution
고정**

현실

**새로운 path 찾으면
seed 추가**

**시간이 지날수록
새 path를 찾을 확률 감소**

A Variant of the Adversarial Multi-Armed Bandit



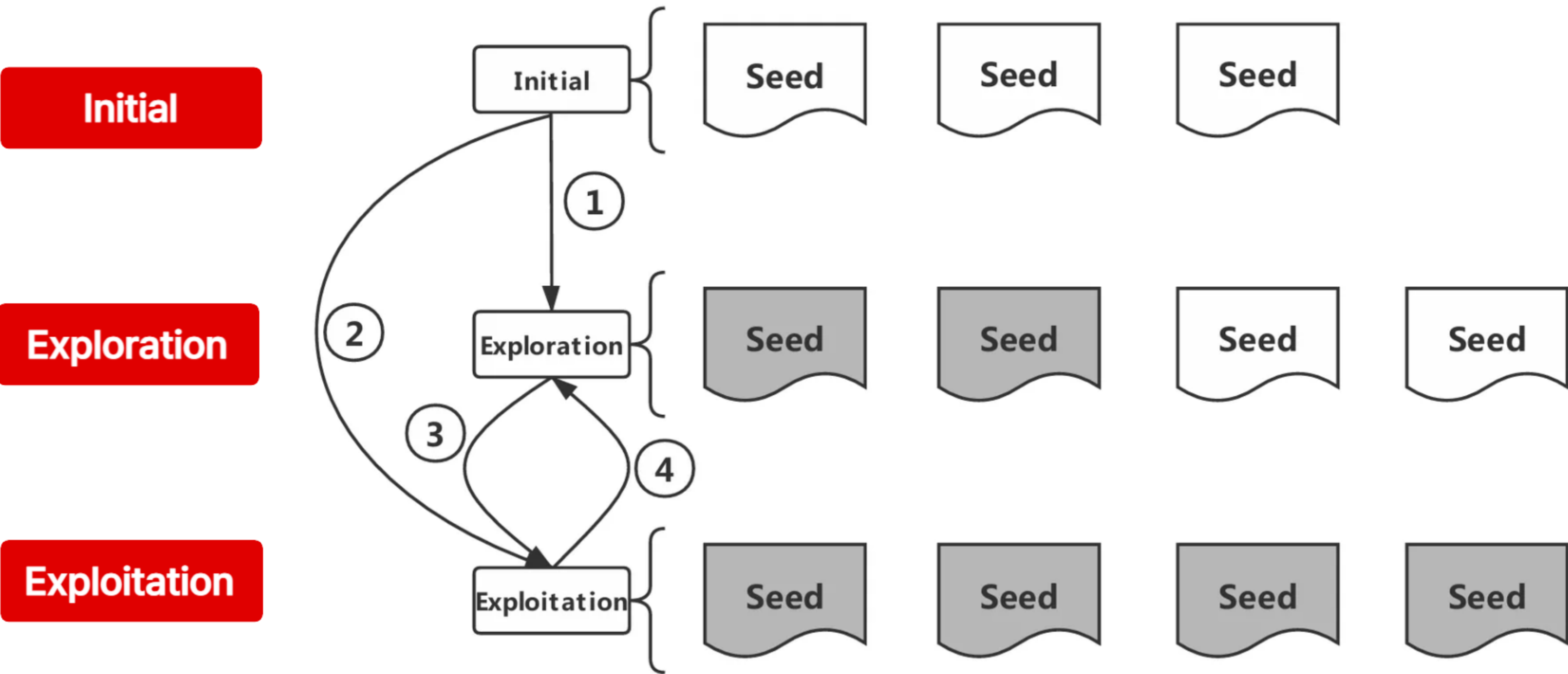
arm $\rightarrow$ seed

trial $\rightarrow$ mutate and execute

reward $\rightarrow$ discovering a new path

Three States of Seed Set

white = unfuzzed seed gray = fuzzed seed



Challenges in VAMAB

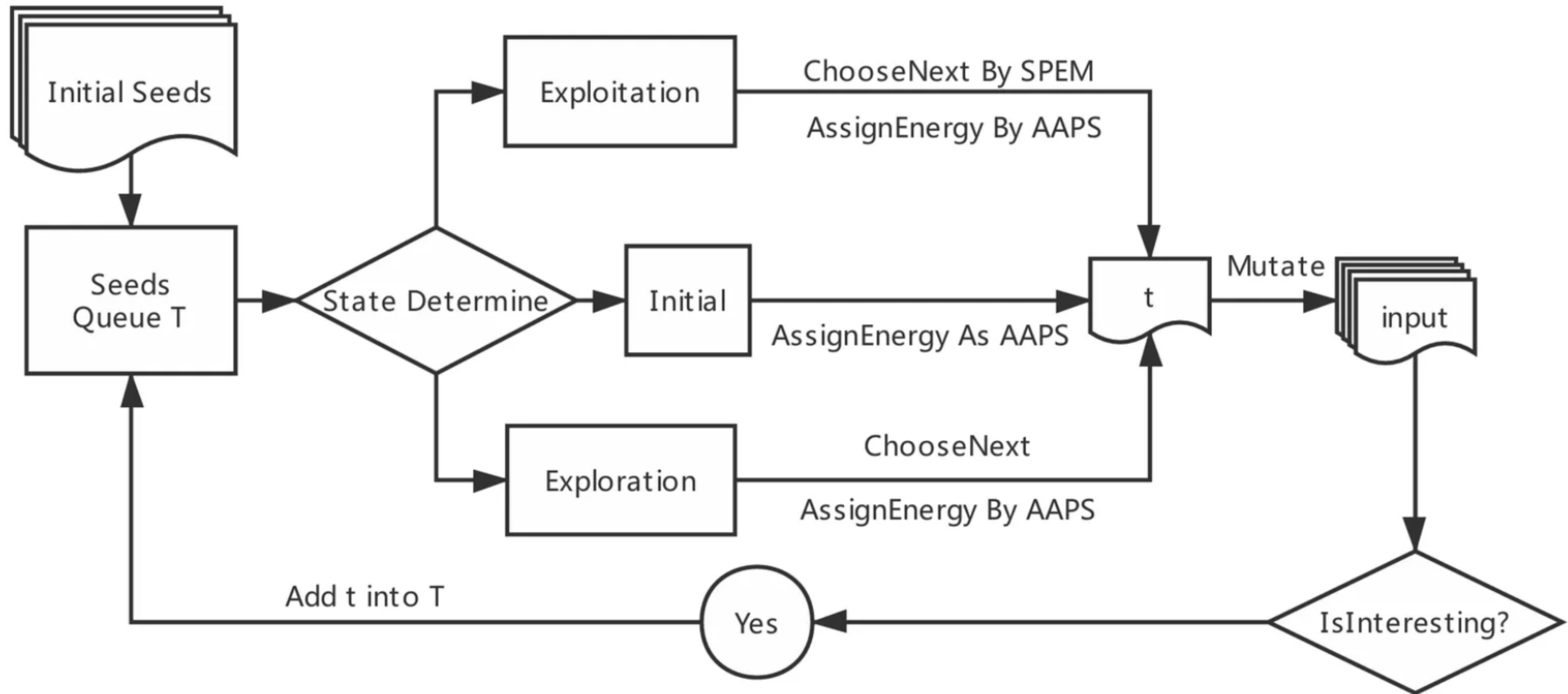
reward probability

hard to estimate accurately

energy allocation

**hard to balance
exploration and exploitation**

Main Framework of EcoFuzz



SPEM: Self-transition-based Probability Estimation Method

$$P_{R_{i,n}} \approx 1 - \frac{f_{ii}}{\sqrt{i}}$$

self-transition frequency ↓

seed index ↑ (later discovered seed)

AAPS: Adaptive Average-Cost-based Power Schedule

exploration

high-frequency path → less energy

low-frequency path → more energy

exploitation

adjust energy using previous allocation result

EcoFuzz